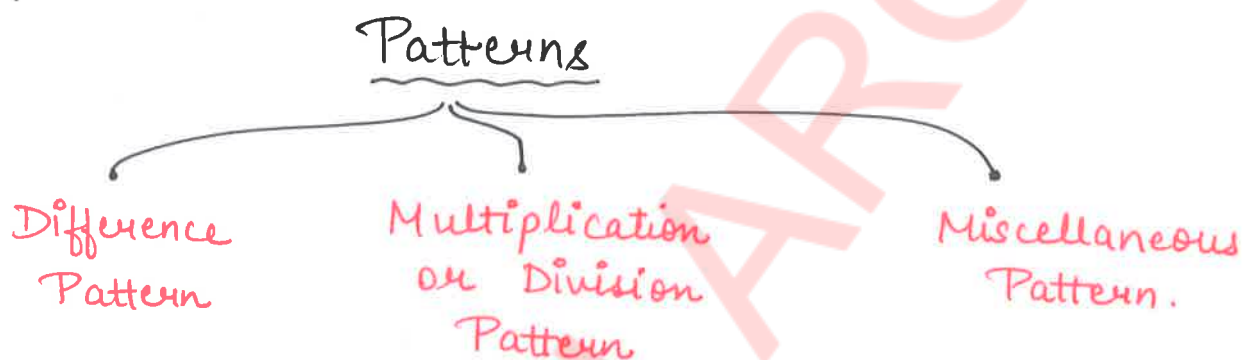


NUMBER SERIES

Missing Number Series
Wrong Number Series

5, 10, 20, 35, 55, 80 → Series
 $+5$ $+10$ $+15$ $+20$ $+25$ → Pattern



★ Missing Number Series :-

⇒ Difference

10, 17, 31, 52, 80, 115
 $+7$ $+14$ $+21$ $+28$ $+35$

⇒ Multiplication

10, 20, 80, 480, 3840
 $\times 2$ $\times 4$ $\times 6$ $\times 8$

Pattern 1 :- Constant Difference (Arithmetic Progression)

Rule :- Add / Subtract a fix number (Common Difference)

Example,

1) 9, 14, 19, 24, 29
 $+5$ $+5$ $+5$ $+5$

2) 50, 45, 40, 35, 30
 -5 -5 -5 -5

Practice Questions:

Q1. 45, 40, 35, 30, 25, ?

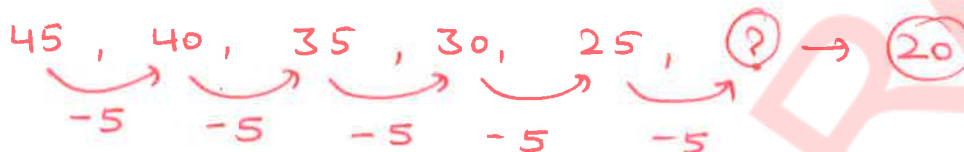
(A) 18

(B) 17

☒ (C) 20

(D) 15

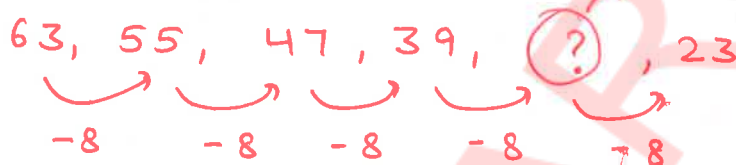
(E) 24



Q2. 63, 55, 47, 39, ?, 23

☒ (A) 31 (B) 33 (C) 35 (D) 29 (E) None of these

$$39 - 8 = 31$$



Q3. ?, 47, 54, 61, 68, 75

(A) 39

☒ (B) 40

(C) 36

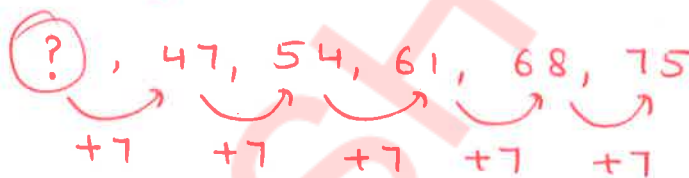
(D) 42

(E) 41

$$x + 7 = 47$$

$$x = 47 - 7$$

$$x = 40$$



Q4. 116, 110, 104, 98, 92, ?

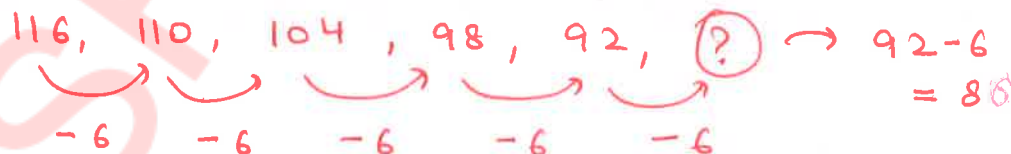
☒ (A) 86

(B) 85

(C) 96

(D) 91

(E) 81



Q5. 96, 105, 114, 123, ?, 141

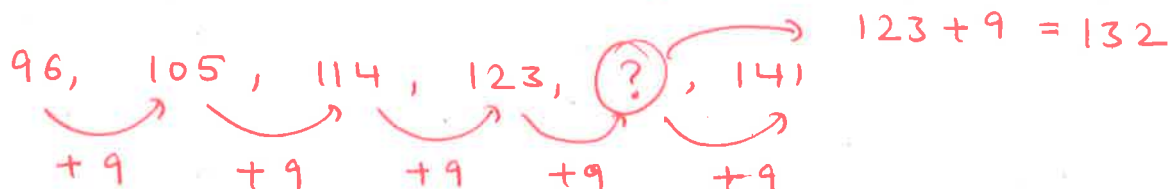
(A) 134

(B) 135

(C) 128

☒ (D) 132

(E) 133



Pattern 2 → Double Difference (Increasing Order)

Rule :- Difference between terms keep increasing in a pattern.

Example,

$$\begin{array}{cccccc}
 6, & 7, & 9, & 12, & 16, & 21 \\
 \swarrow & \swarrow & \swarrow & \swarrow & \swarrow & \\
 +1 & +2 & +3 & +4 & +5 & \\
 \swarrow & \swarrow & \swarrow & \swarrow & \swarrow & \\
 +1 & +1 & +1 & +1 & &
 \end{array}$$

$$\begin{array}{cccccc}
 3, & 5, & 9, & 15, & 23, & 33 \\
 \swarrow & \swarrow & \swarrow & \swarrow & \swarrow & \\
 +2 & +4 & +6 & +8 & +10 & \\
 \swarrow & \swarrow & \swarrow & \swarrow & \swarrow & \\
 +2 & +2 & +2 & +2 & &
 \end{array}$$

Practice Questions:

Q1. 12, 17, 26, 39, 56, ?

(A) 89

(B) 81

(C) 77

(D) 84

(E) 65

$$\begin{array}{cccccc}
 12, & 17, & 26, & 39, & 56, & (?) \\
 \swarrow & \swarrow & \swarrow & \swarrow & \swarrow & \\
 +5 & +9 & +13 & +17 & +21 & \\
 \swarrow & \swarrow & \swarrow & \swarrow & \swarrow & \\
 +4 & +4 & +4 & +4 & &
 \end{array}
 \quad \begin{array}{l}
 56 + 21 \\
 = 77
 \end{array}$$

Q2. 7, 12, 18, 25, ?, 42

(A) 34

(B) 33

(C) 30

(D) 28

(E) 37

$$\begin{array}{cccccc}
 7, & 12, & 18, & 25, & (?) & 42 \\
 \swarrow & \swarrow & \swarrow & \swarrow & \swarrow & \\
 +5 & +6 & +7 & +8 & +9 & \\
 \swarrow & \swarrow & \swarrow & \swarrow & \swarrow & \\
 +1 & +1 & +1 & +1 & &
 \end{array}
 \quad \begin{array}{l}
 25 + 8 \\
 = 33
 \end{array}$$

Q3. 16, 9, 6, 7, ?, 21

(A) 15

(B) 8

(C) 6

(D) 12

(E) 10

$$\begin{array}{cccccc}
 16, & 9, & 6, & 7, & (?) & 21 \\
 \swarrow & \swarrow & \swarrow & \swarrow & \swarrow & \\
 -7 & -3 & +1 & +5 & +9 & \\
 \swarrow & \swarrow & \swarrow & \swarrow & \swarrow & \\
 +4 & +4 & +4 & +4 & &
 \end{array}
 \quad \begin{array}{l}
 7 + 5 \\
 = 12
 \end{array}$$

Pattern 3 → Double Difference (Decreasing Order)

Rule :- Differences between terms keep decreasing in a pattern.

Examples,

• 110, 90, 74, 62, 54, 50

$$\begin{array}{ccccccccc} & \curvearrowright & & \curvearrowright & & \curvearrowright & & \curvearrowright & & \curvearrowright \\ & -20 & & -16 & & -12 & & -8 & & -4 \\ & \curvearrowright & & \curvearrowright & & \curvearrowright & & \curvearrowright & & \\ & +4 & & +4 & & +4 & & +4 & & \end{array}$$

• 90, 61, 39, 24, 16, 15

$$\begin{array}{ccccccccc} & \curvearrowright & & \curvearrowright & & \curvearrowright & & \curvearrowright & & \curvearrowright \\ & -29 & & -22 & & -15 & & -8 & & -1 \\ & \curvearrowright & & \curvearrowright & & \curvearrowright & & \curvearrowright & & \\ & +7 & & +7 & & +7 & & +7 & & \end{array}$$

Practice Questions:

Q1. 125, 104, 87, 74, 65, ?

(A) 56

(B) 62

(C) 64

(D) 53

(E) 60

125, 104, 87, 74, 65, ?

$$\begin{array}{ccccccccc} & \curvearrowright & & \curvearrowright & & \curvearrowright & & \curvearrowright & & \curvearrowright \\ & -21 & & -17 & & -13 & & -9 & & -5 \\ & \curvearrowright & & \curvearrowright & & \curvearrowright & & \curvearrowright & & \\ & +4 & & +4 & & +4 & & +4 & & \end{array}$$

65 - 5 = 60

Q2. ?, 132, 101, 79, 66, 62

(A) 174

(B) 165

(C) 163

(D) 170

(E) 172

← ? , 132, 101, 79, 66, 62

$$\begin{array}{ccccccccc} & \curvearrowright & & \curvearrowright & & \curvearrowright & & \curvearrowright & & \curvearrowright \\ & -40 & & -31 & & -22 & & -13 & & -4 \\ & \curvearrowright & & \curvearrowright & & \curvearrowright & & \curvearrowright & & \\ & +9 & & +9 & & +9 & & +9 & & \end{array}$$

$x - 40 = 132$
 $x = 132 + 40$
 $x = 172$

Q3. 185, 174, 166, 161, 159, ?

(A) 150

(B) 157

(C) 154

(D) 160

(E) 155

$$\begin{array}{ccccccccc}
 185 & , & 174 & , & 166 & , & 161 & , & 159 & , & ? & \rightarrow & 159 + 1 \\
 \downarrow & & \downarrow & & \downarrow & & \downarrow & & \downarrow & & & & = 160 \\
 -11 & & -8 & & -5 & & -2 & & +1 & & & & \\
 \downarrow & & \downarrow & & \downarrow & & \downarrow & & & & & & \\
 +3 & & +3 & & +3 & & +3 & & & & & &
 \end{array}$$

Pattern 4 :- Alternate Differences

Rule :- Two different differences repeat alternatively.

Patterns can be $\rightarrow$

- ① $+a, +b, +a, +b, +a, +b, \dots$
- ② $-a, -b, -a, -b, -a, -b, \dots$
- ③ $+a, -b, +a, -b, +a, -b, \dots$
- ④ $-a, +b, -a, +b, -a, +b, \dots$

Examples,

$$\begin{array}{l}
 \bullet \quad 4, 6, 11, 13, 18, 20, 25 \\
 \quad \downarrow \quad \downarrow \quad \downarrow \quad \downarrow \quad \downarrow \quad \downarrow \\
 \quad +2 \quad +5 \quad +2 \quad +5 \quad +2 \quad +5 \\
 \\
 \bullet \quad 30, 27, 34, 31, 38, 35, ? \rightarrow 42 \\
 \quad \downarrow \quad \downarrow \quad \downarrow \quad \downarrow \quad \downarrow \quad \downarrow \\
 \quad -3 \quad +7 \quad -3 \quad +7 \quad -3 \quad +7 \\
 \\
 \bullet \quad 4, 6, 11, 13, 18, 20, 25, 27 \\
 \quad \downarrow \quad \downarrow \quad \downarrow \quad \downarrow \quad \downarrow \quad \downarrow \quad \downarrow \\
 \quad +2 \quad +5 \quad +2 \quad +5 \quad +2 \quad +5 \quad +2
 \end{array}$$

Practice Questions:

Q1. 38, 43, 37, 42, 36, ?

(A) 42

(B) 41

(C) 36

(D) 40

(E) 38

$$\begin{array}{ccccccccc}
 38 & , & 43 & , & 37 & , & 42 & , & 36 & , & ? & \rightarrow & 36 + 5 \\
 \downarrow & & \downarrow & & \downarrow & & \downarrow & & \downarrow & & & & = 41 \\
 +5 & & -6 & & +5 & & -6 & & +5 & & & &
 \end{array}$$

Q2. 48, 51, 57, 60, ?, 69

(A) 56

(B) 66

(C) 65

(D) 59

(E) 62

$$48, 51, 57, 60, ?, 69$$

$+3 \quad +6 \quad +3 \quad +6 \quad +3$

$60 + 6 = 66$

Q3. ?, 71, 69, 76, 74, 81

(A) 64

(B) 52

(C) 61

(D) 68

(E) 67

$$?, 71, 69, 76, 74, 81$$

$+7 \quad -2 \quad +7 \quad -2 \quad +7$

$x + 7 = 71$
 $x = 71 - 7$
 $x = 64$

Q4. 51, 54, 58, 61, 65, ?

(A) 73

(B) 70

(C) 61

(D) 71

(E) 68

$$51, 54, 58, 61, 65, ?$$

$+3 \quad +4 \quad +3 \quad +4 \quad +3$

$65 + 3 = 68$

Q5. 35, 33, 28, 26, 21, ?

(A) 16

(B) 18

(C) 20

(D) 15

(E) 19

$$35, 33, 28, 26, 21, ?$$

$-2 \quad -5 \quad -2 \quad -5 \quad -2$

$21 - 2 = 19$

Pattern 5 → Prime Differences

Rule :- Differences are consecutive Prime Numbers

Prime Numbers :-

A natural number > 1 , with exactly two distinct positive factors → 1 and itself.

1 - 100 Prime Numbers

{ 2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37, 41, 43, 47, 53, 59, 61, 67, 71, 73, 79, 83, 89, 97 }

Prime Numbers

1 - 100 → 25 prime numbers

1 - 50 → 15 prime numbers

50 - 100 → 10 prime numbers

• Definitions

Even Numbers →

Numbers divisible by 2 → are even Numbers.

like, 10, 12, 4, 6,
8, 10, 12,
-2, -8, -48, -4720,
-8292, ----

Odd Numbers →

Numbers that are not divisible by 2 are odd Numbers.

→ 3, 7, 9, 21, 43,
-81, -433, -8295, ----

Prime Numbers →

Numbers that are divisible by 1 and itself are Prime Numbers.

→ 2, 3, 5, 7, 11,
13, 17, 19, ----

Examples,

- 4, 7, 12, 19, 30, 43, 60
- +3 +5 +7 +11 +13 +17

{Consecutive Prime Numbers}

- 10, 12, 15, 20, 27, 38, 51
- +2 +3 +5 +7 +11 +13

Practice Questions:

Q1. 13, 72, 133, 200, 271, 344, ?

(A) 424

(B) 423

(C) 418

(D) 398

(E) 422

13, 72, 133, 200, 271, 344, ?

+59 +61 +67 +71 +73 +79

344 + 79 = 423

Q2. 13, 32, 55, 84, 115, ?, 193

(A) 147

(B) 153

(C) 152

(D) 145

(E) 151

13, 32, 55, 84, 115, ?, 193

+19 +23 +29 +31 +37 +41

115 + 37 = 152

Q3. ?, 30, 43, 60, 79, 102, 131

(A) 24

(B) 19

(C) 14

(D) 22

(E) 20

?, 30, 43, 60, 79, 102, 131

+11 +13 +17 +19 +23 +29

$x + 11 = 30$
 $x = 30 - 11$
 $x = 19$

Pattern 6 → Square Differences

Rule :- Differences are consecutive square numbers
(1, 4, 9, 16, 25, ...)

Examples,

- $2, 3, 7, 16, 32, 57$
 $\swarrow \quad \swarrow \quad \swarrow \quad \swarrow \quad \swarrow$
 $+1 \quad +4 \quad +9 \quad +16 \quad +25$
 $(1^2) \quad (2^2) \quad (3^2) \quad (4^2) \quad (5^2)$
- $10, 14, 23, 29, 64, 100$
 $\swarrow \quad \swarrow \quad \swarrow \quad \swarrow \quad \swarrow$
 $+4 \quad +9 \quad +16 \quad +25 \quad +36$
 $(2^2) \quad (3^2) \quad (4^2) \quad (5^2) \quad (6^2)$

Practice Questions:

Q1. 19, 23, 32, 48, 73, ?

(A) 124

(B) 111

(C) 110

(D) 109

(E) 112

$$\begin{array}{ccccccccc}
 19 & 23 & 32 & 48 & 73 & ? & & & \\
 \swarrow & \swarrow & \swarrow & \swarrow & \swarrow & \swarrow & & & \\
 +4 & +9 & +16 & +25 & +36 & & & & \\
 (2^2) & (3^2) & (4^2) & (5^2) & (6^2) & & & &
 \end{array}$$

$73 + 36 = 109$

Q2. 194, ?, 81, 45, 20, 4

(A) 131

(B) 132

(C) 137

(D) 134

(E) 130

$$\begin{array}{ccccccccc}
 194 & ? & 81 & 45 & 20 & 4 & & & \\
 \swarrow & \swarrow & \swarrow & \swarrow & \swarrow & \swarrow & & & \\
 -64 & -49 & -36 & -25 & -16 & & & & \\
 (8^2) & (7^2) & (6^2) & (5^2) & (4^2) & & & &
 \end{array}$$

$194 - 64 = 130$

Q3. 120, 156, 220, 320, 464, 660

(A) 448

(B) 452

(C) 460

(D) 464

(E) 468

$$\begin{array}{ccccccccc}
 ? & 120 & 156 & 220 & 320 & 464 & 660 & & \\
 \swarrow & \swarrow & \swarrow & \swarrow & \swarrow & \swarrow & \swarrow & & \\
 +16 & +36 & +64 & +100 & +144 & +196 & & & \\
 (4^2) & (6^2) & (8^2) & (10^2) & (12^2) & (14^2) & & &
 \end{array}$$

$x + 16 = 120$
 $x = 120 - 16$
 $x = 104$

(F) 104

Pattern 7 :- Cube Differences

Rule :- Differences are consecutive Cube Numbers.
(1, 8, 27, 64, 125, ...)

Examples,

$$\begin{array}{ccccccccc}
 3, & 4, & 12, & 39, & 103, & 228 \\
 \swarrow & \swarrow & \swarrow & \swarrow & \swarrow \\
 +1 & +8 & +27 & +64 & +125 \\
 (1^3) & (2^3) & (3^3) & (4^3) & (5^3)
 \end{array}$$

$$\begin{array}{ccccccccc}
 5, & 13, & 40, & 104, & 229, & 445 \\
 \swarrow & \swarrow & \swarrow & \swarrow & \swarrow \\
 +8 & +27 & +64 & +125 & +216 \\
 (2^3) & (3^3) & (4^3) & (5^3) & (6^3)
 \end{array}$$

Practice Questions:

Q1. 19, 27, 54, 118, 243, ?

(A) 456

(B) 459

(C) 471

(D) 439

(E) 460

$$\begin{array}{ccccccccc}
 19, & 27, & 54, & 118, & 243, & ? \\
 \swarrow & \swarrow & \swarrow & \swarrow & \swarrow \\
 +8 & +27 & +64 & +125 & +216 \\
 (2^3) & (3^3) & (4^3) & (5^3) & (6^3)
 \end{array}$$

$243 + 216 = 459$

Q2. 2000, 1488, 1145, 929, 804, ?

(A) 710

(B) 740

(C) 720

(D) 750

(E) 730

$$\begin{array}{ccccccccc}
 2000, & 1488, & 1145, & 929, & 804, & ? \\
 \swarrow & \swarrow & \swarrow & \swarrow & \swarrow \\
 -512 & -343 & -216 & -125 & -64 \\
 (8^3) & (7^3) & (6^3) & (5^3) & (4^3)
 \end{array}$$

$804 - 64 = 740$

Q3. 50, 58, 122, 338, ?, 1850

(A) 720

(B) 850

(C) 1040

(D) 790

(E) 910

$$\begin{array}{ccccccccc}
 50, & 58, & 122, & 338, & ? & 1850 \\
 \swarrow & \swarrow & \swarrow & \swarrow & \swarrow \\
 +8 & +64 & +216 & +512 & +1000 \\
 (2^3) & (4^3) & (6^3) & (8^3) & (10^3)
 \end{array}$$

$338 + 512 = 850$

Pattern 8 → Alternate Square Difference

Rule:- Square differences with alternate sign (+1-)

Examples,

$$\begin{array}{cccccc}
 20, & 24, & 15, & 31, & 6, & 42 \\
 \curvearrowright & \curvearrowright & \curvearrowright & \curvearrowright & \curvearrowright & \\
 +4 & -9 & +16 & -25 & +36 & \\
 (2^2) & (3^2) & (4^2) & (5^2) & (6^2) &
 \end{array}$$

$$\begin{array}{cccccc}
 40, & 49, & 33, & 58, & 22, & 71 \\
 \curvearrowright & \curvearrowright & \curvearrowright & \curvearrowright & \curvearrowright & \\
 +9 & -16 & +25 & -36 & +49 & \\
 (3^2) & (4^2) & (5^2) & (6^2) & (7^2) &
 \end{array}$$

Practice Questions:

Q1. 120, ?, 113, 138, 102, 151

(A) 129

(B) 124

(C) 136

(D) 121

(E) 125

$$\begin{array}{cccccc}
 120, & ?, & 113, & 138, & 102, & 151 \\
 \curvearrowright & \curvearrowright & \curvearrowright & \curvearrowright & \curvearrowright & \\
 +9 & -16 & +25 & -36 & +49 & \\
 + (3^2) & - (4^2) & + (5^2) & - (6^2) & + (7^2) &
 \end{array}$$

120 + 9 = 129

Q2. 300, ?, 275, 471, 246, 502

(A) 452

(B) 456

(C) 441

(D) 444

(E) 463

$$\begin{array}{cccccc}
 300, & ?, & 275, & 471, & 246, & 502 \\
 \curvearrowright & \curvearrowright & \curvearrowright & \curvearrowright & \curvearrowright & \\
 +144 & -169 & +196 & -225 & +256 & \\
 (12^2) & - (13^2) & + (14^2) & - (15^2) & + (16^2) &
 \end{array}$$

300 + 144 = 444

Q3. 200, ?, 184, 233, 152, 273

(A) 207

(B) 209

(C) 211

(D) 213

(E) 205

$$\begin{array}{cccccc}
 200, & ?, & 184, & 233, & 152, & 273 \\
 \curvearrowright & \curvearrowright & \curvearrowright & \curvearrowright & \curvearrowright & \\
 +9 & -25 & +49 & -81 & +121 & \\
 + (3^2) & - (5^2) & + (7^2) & - (9^2) & + (11^2) &
 \end{array}$$

200 + 9 = 209

Pattern 9 → Alternate Cube Difference

Rule :- Cube Difference with Alternate Sign (+/-)

Example,

- $50, 51, 43, 70, 6, ?$
 $+1 \quad -8 \quad +27 \quad -64 \quad +125$
 $+(1^3) \quad -(2^3) \quad +(3^3) \quad -(4^3) \quad +(5^3)$
- $120, 128, 101, 165, 40, 256$
 $+8 \quad -27 \quad +64 \quad -125 \quad +216$
 $+(2^3) \quad -(3^3) \quad +(4^3) \quad -(5^3) \quad +(6^3)$

Practice Questions:

Q1. 300, ?, 281, 345, 220, 436

(A) 325

(B) 312

(C) 316

(D) 292

(E) 308

$300 + 8 = 308$
 $300, ?, 281, 345, 220, 436$
 $+8 \quad -27 \quad +64 \quad -125 \quad +216$
 $(2^3) \quad (3^3) \quad (4^3) \quad (5^3) \quad (6^3)$

Q2. 1000, ?, 909, 1252, 740, 1469

(A) 1008

(B) 1064

(C) 1089

(D) 1125

(E)

1144

$1000 + 125 = 1125$
 $1000, ?, 909, 1252, 740, 1469$
 $+125 \quad -216 \quad +343 \quad -512 \quad +729$
 $(5^3) \quad (6^3) \quad (7^3) \quad (8^3) \quad (9^3)$

Q3. 500, 527, 402, 745, 16, ?

(A) 1260

(B) 1347

(C) 1296

(D) 1300

(E) 1325

$16 + 1331 = 1347$
 $500, 527, 402, 745, 16, ?$
 $+27 \quad -125 \quad +343 \quad -729 \quad +1331$
 $(3^3) \quad (5^3) \quad (7^3) \quad (9^3) \quad (11^3)$

Pattern 10 :- Cubic Series

Rule :- Based on n^3 or $n^3 \pm n$ formulas.

Examples,

$$(n^3 + n) \rightarrow 2, 10, 30, 68, 130, 222, \dots$$

$$\begin{array}{cccccc} \downarrow & \downarrow & \downarrow & \downarrow & \downarrow & \downarrow \\ (1+1) & (8+2) & (27+3) & (64+4) & (125+5) & (216+6) \\ (1^3+1) & (2^3+2) & (3^3+3) & (4^3+4) & (5^3+5) & (6^3+6) \end{array}$$

$$n(n+1) \rightarrow 1 \times 2, 2 \times 3, 3 \times 4, 4 \times 5, 5 \times 6, 6 \times 7, \dots$$

$$\begin{array}{cccccc} 2 & 6 & 12 & 20 & 30 & 42 \end{array}$$

Practice Questions:

Q1. 28, 65, 126, 217, ?, 513

(A) 384

(B) 344

(C) 351

(D) 342

(E) 337

$$\begin{array}{cccccc} 28, & 65, & 126, & 217, & (?), & 513 \\ \downarrow & \downarrow & \downarrow & \downarrow & \downarrow & \downarrow \\ (3^3+1) & (4^3+1) & (5^3+1) & (6^3+1) & (7^3+1) & (8^3+1) \\ & & & & \hookrightarrow 343+1 = 344 \end{array}$$

Q2. 6, 13, 32, 69, 130, ?

(A) 221

(B) 196

(C) 261

(D) 217

(E) 233

$$\begin{array}{cccccc} 6, & 13, & 32, & 69, & 130, & (?), \rightarrow 221 \\ \downarrow & \downarrow & \downarrow & \downarrow & \downarrow & \downarrow \\ (1+5) & (8+5) & (27+5) & (64+5) & (125+5) & (216+5) \\ (1^3+5) & (2^3+5) & (3^3+5) & (4^3+5) & (5^3+5) & (6^3+5) \end{array}$$